

Subspace Alignment for CLIP-based Continual Learning via Canonical Correlation Analysis

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Abstract

Recent advances in CLIP-based continual learning have shown the potential of leveraging pre-trained vision-language models for sequential tasks. However, existing methods overlook a key problem we call *Asymmetric Drift*. In unimodal CLIP-based continual learning, the visual branch undergoes stronger adaptation because the visual distribution shifts significantly, whereas the text branch remains relatively stable due to the low variance of textual prompts. This imbalance increases the modality distance and degrades cross-modal alignment over time. To address this issue, we propose CCA-CL, a framework that accumulates visual-textual covariance statistics across tasks and solves Canonical Correlation Analysis to compute a shared subspace. In this subspace, the distance between visual and textual features is minimized, enabling better alignment without modifying CLIP parameters. This also makes our method naturally compatible with exemplar-free CL settings. To further capture nonlinear relationships that linear Canonical Correlation Analysis is hard to model, we introduce Random Fourier Projection as an extension. Experimental results demonstrate that CCA-CL effectively mitigates the asymmetric drift problem and achieves state-of-the-art performance on several benchmarks. We release the code at <https://github.com/zhwhu/CCA-CL>.

1. Introduction

Continual learning (CL) enables models to learn from a sequence of tasks while preserving previously acquired knowledge. Traditional approaches often train models from scratch [5], which easily causes catastrophic forgetting of past knowledge [11]. With the rise of large pre-trained models [26, 47], recent studies increasingly rely on frozen pre-trained models as feature extractors to enhance stability and

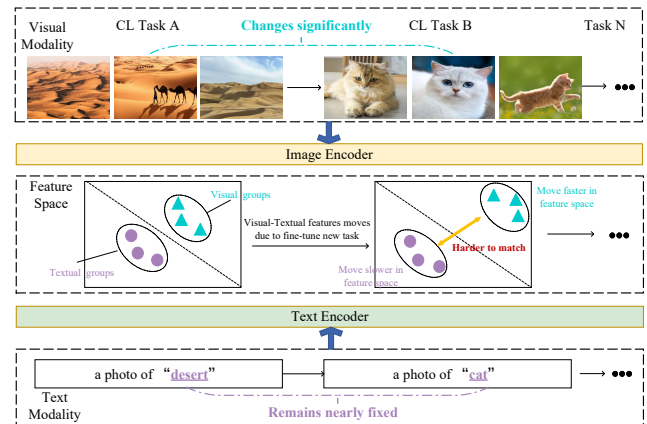


Figure 1. Asymmetric Drift in CLIP-based continual learning. Due to larger updates from high visual variability, the image encoder shifts more than the text encoder, causing visual features to drift faster in the feature space. This imbalance increases the distance between visual and textual features, making cross-modal alignment harder over time.

transferability. Among them, vision-language models such as CLIP [35] show strong generalization and zero-shot capability across diverse domains such as remote sensing [9], industrial safety monitoring [1], plant disease detection [3], and so on. These advantages also make CLIP an efficient continual learner, motivating researchers to explore how to adapt CLIP to continual tasks.

Existing CLIP-based CL methods can be roughly categorized into two groups. The first group freezes CLIP as a fixed feature extractor and trains additional modules to adapt its representations to new tasks [16, 49]. The second group fine-tunes CLIP itself or using parameter-efficient techniques to enhance adaptability while keeping the pre-trained backbone largely intact [39, 46]. Overall, both paradigms aim to preserve the original CLIP representations, focusing either on improving feature fusion [17] for better task performance or on leveraging textual guidance to refine visual adaptation [25, 48]. Despite their architec-

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tural differences, both paradigms adapt CLIP to continual tasks by training a subset of parameters, whether in the base model [43], additional task-specific modules [48], or PEFT components [46]. However, they overlook a critical issue of Asymmetric Drift (AD) during continual updates. Specifically, in unimodal CLIP-based CL, the visual branch undergoes substantially larger updates due to high intra-task distributional shifts across visual inputs. In contrast, the text is mainly composed of prompts and discrete token embeddings and the text branch receives much smaller updates due to the lower variance in text distributions. As illustrated in Figure 1, from Task A (desert scenes) to Task B (cat images), the visual inputs change significantly, leading to large shifts in the visual encoder. However, the corresponding textual descriptions vary only slightly—shifting from “a photo of desert” to “a photo of cat”. This contrast in data variability leads to an update imbalance between the visual and textual encoders. This update imbalance causes the visual and textual features to shift at different rates in the feature space, resulting in increasing modality distance and making visual-textual features harder to match. Over time, this mismatch degrades cross-modal correspondence, making it increasingly difficult for CLIP to maintain reliable vision-language grounding in CL scenarios.

In this paper, we aim to address the AD Problem in CLIP-based CL. AD arises from asymmetric updates to the visual and textual encoders across tasks, causing their feature representations to diverge in the feature space. A natural solution is to reduce the distance between visual and textual features. However, reducing this distance within CLIP’s original feature space typically requires aggressive fine-tuning of CLIP itself, which carries the risk of destabilizing the pre-trained knowledge and compromising its generalization capability. To avoid this, we propose a Canonical Correlation Analysis-based Continual Learning (CCA-CL) framework to address these challenges. Canonical Correlation Analysis (CCA) is a classical statistical method introduced in 1936 [15], which can find linear projections that maximize the correlation between two sets of variables. In practice, CCA-CL accumulates cross-task covariance statistics and solves the CCA equations to compute a shared latent subspace. In this subspace, the distance between visual and textual features is minimized, making them easier to align. By progressively aligning the modalities in this space after each task, our method effectively compensates for asymmetric drift and preserves the consistency of visual-textual representations. Since our approach accumulates only covariance matrices, it implicitly retains information about past data distributions without relying on exemplars, making it naturally suitable for exemplar-free CL. Furthermore, since CCA is limited to modeling linear relationships, its expressiveness may restrict performance in complex CL scenarios. To address this limitation, we ex-

tend CCA with Random Fourier Projection (RFP) to capture nonlinear dependencies between visual and textual features that linear CCA cannot model. Thanks to its lightweight design, CCA-CL enables efficient CL while achieving SOTA performance on several benchmarks.

The main contributions of this paper are as follows:

1. We propose Canonical Correlation Analysis-based Continual Learning framework, which uses canonical correlation analysis to solve the Asymmetric Drift problem by computing a shared subspace that maximizes the canonical correlation between visual and textual features.
2. We propose to use Random Fourier Projection to extend canonical correlation analysis for capturing nonlinear visual-textual correlations for better alignment.
3. We conduct extensive experiments on four datasets, and the empirical results consistently demonstrate the SOTA performance and high efficiency of our method.

2. Related work

2.1. Continual Learning

Traditional CL methods can be categorized into three categories. 1) Regularization-based methods [4, 22, 23] use historical knowledge to consolidate past knowledge by extending the loss function with an additional regularization term. 2) Architecture-based methods dynamically allocate model parameters to specific tasks [19, 20, 41]. These methods either extend the model with new parameters as tasks increase or freeze certain parameters to reduce forgetting. 3) Replay-based methods maintain a memory buffer or synthesize pseudo-samples to revisit past experiences [6, 14, 30, 32]. Although replay-based methods effectively alleviate forgetting, they require additional memory for saving past samples. Recently, with the rapid development of pre-trained models, researchers have begun to explore how to adapt them for continual tasks. SLCA++[44] sequentially fine-tunes low-rank LoRA matrices with a small learning rate to avoid disrupting the pre-trained representations. RanPAC [26] exploits random projectors and class-prototype accumulation thus enhancing the pre-trained representations. APER [47] combines the adapted pre-trained models with the original models for balancing generalization and task-specific learning.

2.2. Continual Learning on CLIP

Vision-language models such as CLIP demonstrate superior zero-shot performance compared to other large models, making it particularly suitable for CL scenarios. Continual-CLIP [35] demonstrates that even without any fine-tuning or additional modules, CLIP can achieve reasonable performance in CL. For this reason, researchers have begun to explore more efficient ways to leverage CLIP’s representations for continual task adaptation in recent years. One line

of work freezes CLIP and introduces additional learnable modules to project CLIP features for continual-task adaptation [16]. PROOF [49] integrates learnable modules with CLIP’s output features to facilitate cross-modal interaction. ENGINE [48] uses incremental linear units to inject additional textual knowledge. Another line of work adapts CLIP to continual tasks by fine-tuning the model itself in different ways [24, 39]. Magmax [25] sequentially fine-tunes the CLIP and creates task vectors by subtracting the weights of the model. LGVLM [46] trains separate LoRA modules for each task. A third line of work investigate CLIP-based CL from the perspective of modal misalignment. MG-CLIP [17] addresses the modality gap by preserving a stable gap for stability while compensating for its limitations to improve plasticity. Mod-X [27] analyzes modal relations from a geometric viewpoint, focusing on intra-modal rotation and inter-modal deviation. Different from the methods mentioned above, we analyze the AD problem and formulate it from a distance-based perspective. Based on this formulation, we use CCA to project visual and textual features into a shared subspace, thereby enhancing their alignment.

2.3. Canonical Correlation Analysis

CCA was first introduced by Hotelling [15] in 1936 and is a classical multivariate statistical method used to maximize the correlation between linear combinations of variables across datasets. Modern extensions such as multiset CCA [28], kernel CCA [10], Deep CCA [2], and sparse CCA [12] have significantly enhanced the modeling power of CCA, especially in handling nonlinear relationships, hierarchical structures, and high-dimensional data. These methods have extended the application of CCA to various domains, such as neuroscience [50], biochemistry [31] and computer science [40]. In the field of machine learning, CCA also has applications. For example, Sun [34] employs Deep CCA to correlate representations across text, audio, and video modalities. Zhang [45] proposes a self-supervised graph representation learning method inspired by classical CCA but optimized at the feature level rather than the instance level. Chen [7] presents a K-means clustering-based kernel CCA algorithm for multimodal emotion recognition in human–robot interaction. In this paper, we introduce CCA into CLIP-based CL. Unlike deep or kernelized variants that require training, our approach leverages the closed-form nature of CCA to perform efficient statistical alignment between visual and textual features across continual tasks.

3. Preliminaries

3.1. Problem Definition

In this work, we focus on class-incremental learning (CIL) with a pre-trained CLIP model F which consists of a vi-

sual encoder $F_v(\cdot)$ and a textual encoder $F_t(\cdot)$. The goal is to sequentially adapt the model on a series of tasks $T = \{D_1, \dots, D_T\}$, where each task t consists of a disjoint set of categories C_t , and the categories between tasks do not overlap, i.e., $C_i \cap C_j = \emptyset$ for $i \neq j$. During each task t , the model receives image inputs $x \in \mathcal{X}_t$ and their class labels $y \in C_t$. For each class c , its textual prototype $z_y = F_t(\text{“a photo of [CLS]”})$ is generated from a hand-crafted prompt such as “a photo of [CLS]”. At inference time, the classification is performed by computing the cosine similarity between the visual embedding $z_x = F_v(x)$ and all textual prototypes z_y^c , and assigning the label corresponding to the highest similarity:

$$\hat{y} = \arg \max_{c \in C_1 \cup \dots \cup C_t} \text{sim}(z_x, z_y^c), \quad (1)$$

where

$$\text{sim}(z_x, z_y^c) = \frac{z_x^\top z_y^c}{\|z_x\|_2 \|z_y^c\|_2}. \quad (2)$$

Intuitively, the image distribution continuously shifts as new tasks arrive, whereas the textual features remain almost static. This leads to asymmetric updates between the visual and textual encoders. In this paper, we employ CCA to mitigate this issue.

3.2. Preliminaries on CCA

Given two sets of random variables X (e.g image features) and Y (e.g text features), CCA finds two projection matrices A and B that maximize the correlation between the projected variables $A^\top X$ and $B^\top Y$:

$$\max_{A, B} \text{corr}(A^\top X, B^\top Y) = \frac{A^\top \Sigma_{XY} B}{\sqrt{A^\top \Sigma_{XX} A} \sqrt{B^\top \Sigma_{YY} B}}, \quad (3)$$

where Σ_{XX} , Σ_{YY} , and Σ_{XY} are the covariance matrices of X , Y , and their cross-covariance. This optimization can be equivalently solved via the eigenvalue decomposition of the whitened cross-covariance:

$$K = \Sigma_{XX}^{-\frac{1}{2}} \Sigma_{XY} \Sigma_{YY}^{-\frac{1}{2}} = V \Lambda U^\top, \quad (4)$$

where Λ is a diagonal matrix containing the canonical correlations, and V and U denote the canonical directions for X and Y , respectively. Larger diagonal entries of Λ indicate stronger linear dependencies between the corresponding projected directions, meaning that the two modalities are highly aligned along these axes. Hence, the canonical directions define the most correlated subspaces between X and Y . The resulting projections are given by:

$$A = \Sigma_{XX}^{-\frac{1}{2}} V, \quad B = \Sigma_{YY}^{-\frac{1}{2}} U, \quad (5)$$

which project both modalities into a shared subspace where their correlation is maximized. Through CCA, the most statistically correlated subspaces between the two modalities

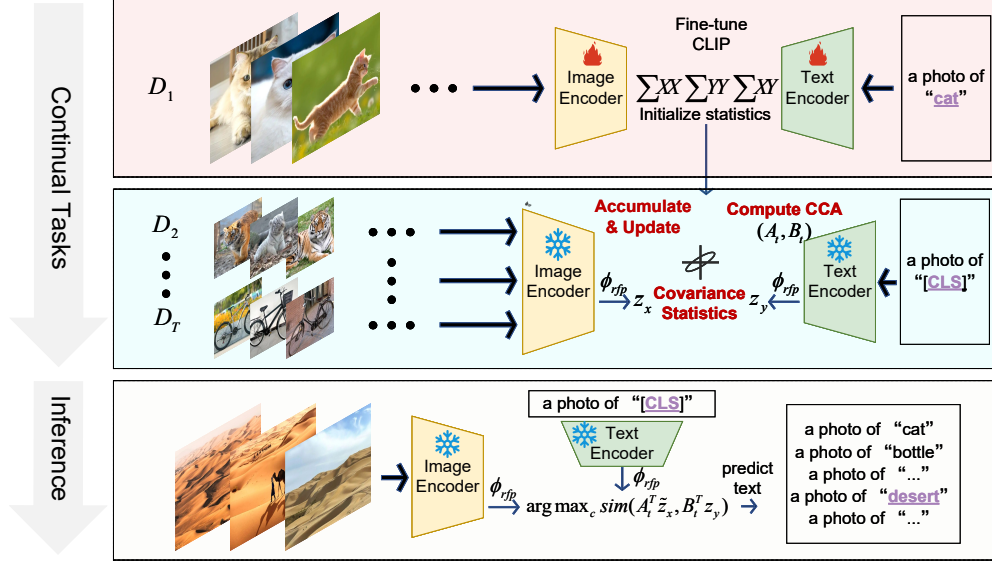


Figure 2. Overview of CCA-CL. Given a continual task stream $\{D_1, \dots, D_T\}$, we optionally fine-tune CLIP on the first task and then freeze the CLIP. For each subsequent task, visual and textual features are extracted (optionally mapped by RFP), and cross-task covariance statistics ($\Sigma_{xx}, \Sigma_{yy}, \Sigma_{xy}$) are accumulated. The projection matrices (A_t, B_t) are then obtained by solving a closed-form CCA problem. During inference, image features and text prototypes are projected into the shared CCA subspace and classified via cosine similarity.

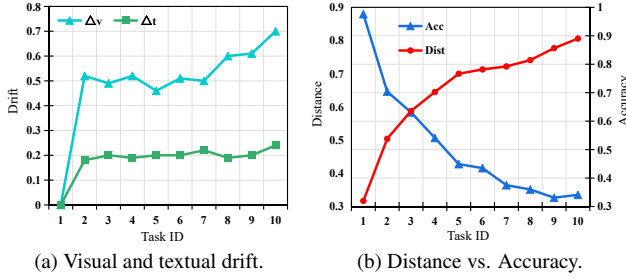


Figure 3. Drift and modality misalignment in CLIP-based CL. (a) The visual drift Δ_v consistently exceeds the textual drift Δ_t , indicating asymmetric drift. (b) This asymmetry causes the visual-textual feature distance to increase over time, leading to degraded accuracy.

can be effectively discovered, revealing the intrinsic linear dependencies that serve as the foundation for cross-modal alignment.

4. Method

4.1. Asymmetric Drift in CLIP-based CL

We define the visual drift as:

$$\Delta_v = \|\mathbb{E}_{x \in \mathcal{D}_t}[z_x] - \mathbb{E}_{x \in \mathcal{D}_{t-1}}[z_x]\|_2, \quad (6)$$

which quantifies the change in the mean visual features between the current task and the previous task. Similarly, the textual drift is defined as:

$$\Delta_t = \|\mathbb{E}_{y \in \mathcal{D}_t}[z_y] - \mathbb{E}_{y \in \mathcal{D}_{t-1}}[z_y]\|_2, \quad (7)$$

which measures the change in the mean textual features across two consecutive tasks. What’s more, we also quantify the visual-textual feature distance at task t as:

$$\text{dist}(t) = \mathbb{E}_{(x,y) \in \mathcal{D}_t} [\|z_x^{(t)} - z_y^{(t)}\|_2], \quad (8)$$

which measures the average ℓ_2 distance between visual and textual features over all image-text pairs in task t . A larger distance indicates greater misalignment caused by AD. We track Δ_v , Δ_t , and $\text{dist}(t)$ throughout the CL process by fine-tuning CLIP over a sequence of tasks. As shown in Figure 3(a), the visual drift Δ_v consistently exceeds the textual drift Δ_t across all tasks. This imbalance leads to an increasing visual-textual feature distance, as illustrated in Figure 3(b), where $\text{dist}(t)$ steadily rises with task progression. This rising modality distance correlates with a significant drop in accuracy, indicating degraded vision-language alignment.

To address the AD problem, a natural solution is to reduce the modality distance at each task, thereby encouraging better alignment between visual and textual features. However, directly minimizing this alignment loss by updating CLIP’s parameters may cause catastrophic forgetting or destroy the pre-trained representations [47]. Thus, a desirable alternative is to project both modalities into a shared subspace where their alignment is maximized—without modifying CLIP itself. This motivates us to consider CCA, which aims to find a subspace in which the distance between visual and textual features is minimized.

4.2. CCA-based Subspace Projection

Given a mini-batch of image features $z_x \in \mathbb{R}^{N \times D}$ extracted by the visual encoder F_v , and the corresponding text features $z_y \in \mathbb{R}^{N \times D}$ obtained from the textual encoder F_t via class prompts, where N is the batch size and D is the feature dimension, we recursively update the covariance statistics as follows:

$$\begin{aligned}\Sigma_{xx} &\leftarrow \Sigma_{xx} + (z_x - \bar{z}_x)^\top (z_x - \bar{z}_x), \\ \Sigma_{yy} &\leftarrow \Sigma_{yy} + (z_y - \bar{z}_y)^\top (z_y - \bar{z}_y), \\ \Sigma_{xy} &\leftarrow \Sigma_{xy} + (z_x - \bar{z}_x)^\top (z_y - \bar{z}_y),\end{aligned}\quad (9)$$

where $\bar{z}_x$ and $\bar{z}_y$ are running means. This running update allows the model to retain long-term cross-task statistical memory while smoothly adapting to new task distributions. After accumulating the covariance statistics, we solve a closed-form CCA to align the two modalities. Specifically, the covariance matrices are whitened via eigen decomposition:

$$\Sigma_{xx} = U_x \Lambda_x U_x^\top, \quad \Sigma_{yy} = U_y \Lambda_y U_y^\top, \quad (10)$$

and the whitening transformations are given by:

$$W_x = U_x \Lambda_x^{-1/2} U_x^\top, \quad W_y = U_y \Lambda_y^{-1/2} U_y^\top. \quad (11)$$

The cross-covariance matrix is then computed as:

$$C = W_x \Sigma_{xy} W_y. \quad (12)$$

We perform eigen decomposition on $K = CC^\top$ to obtain the canonical bases V and canonical correlations Λ_ρ :

$$KV = V\Lambda_\rho, \quad \text{where } \Lambda_\rho = \text{diag}(\rho_1, \dots, \rho_D). \quad (13)$$

The top- r components are then selected according to an energy criterion:

$$\frac{\sum_{i=1}^r \rho_i^2}{\sum_{i=1}^D \rho_i^2} \geq \eta, \quad (14)$$

where η (typically 0.99) preserves most of the total correlation energy while discarding noise-dominated components. Accordingly, the top- r canonical directions are retained, i.e., $V^r = V[:, :r]$ and $\Lambda_\rho^r = \Lambda_\rho[:, :r]$, which are then used to construct the CCA projection matrices. Finally, the CCA projection matrices are derived as:

$$A = W_x V^r, \quad B = W_y (C^\top V_r) \Lambda_\rho^{r-1}. \quad (15)$$

They project the image and text features into a shared correlated subspace:

$$z_x^{cca} = A^\top z_x, \quad z_y^{cca} = B^\top z_y. \quad (16)$$

In summary, we accumulate cross-task covariance statistics and solve a closed-form CCA to obtain projection matrices. These matrices define a shared subspace in which visual and textual features are maximally correlated. This subspace effectively minimizes the distance between the two modalities, allowing better visual-textual feature matching across continual tasks.

Algorithm 1 CCA-based Continual Learning

Require: Continual task stream $\{\mathcal{D}_1, \dots, \mathcal{D}_T\}$, CLIP model F , random feature mapper ϕ_{rfp}
Initialize covariance matrices $\Sigma_{x'x'}, \Sigma_{y'y'}, \Sigma_{x'y'} \leftarrow 0$
for task $t = 1$ to T **do**
 if $t = 1$ **then**
 Fine-tune CLIP on $\mathcal{D}_1$ and then freeze parameters
 end if
 for each mini-batch (x, y) from $\mathcal{D}_t$ **do**
 Image features $z_x = \phi_{\text{rfp}}(F_v(x))$
 Text features $z_y = \phi_{\text{rfp}}(F_t(\text{"a photo of y"}))$
 Update covariance matrices as Eq. 9
 end for
 Compute CCA projections as Eqs. (10)–(16)
end for
Inference: Inference any image x as Eq. 19

4.3. Nonlinear Extension via Random Fourier Projection

Although CCA effectively aligns linear dependencies between the visual and textual representations, it remains limited in capturing complex nonlinear relationships. To enhance this, we introduce a nonlinear extension using Random Fourier Projection (RFP), which approximates a shift-invariant kernel mapping through random feature expansion.

Specifically, during the covariance accumulation stage described in Eq. 9, we apply a random feature mapping $\phi(\cdot)$ to both image and text features before updating their covariance statistics:

$$z'_x = \phi(z_x) = \sqrt{\frac{2}{D'}} \cos(W z_x + b), \quad (17)$$

$$z'_y = \phi(z_y) = \sqrt{\frac{2}{D'}} \cos(W z_y + b), \quad (18)$$

where $W \in \mathbb{R}^{D' \times D}$ are random projection matrices sampled from a Gaussian distribution $\mathcal{N}(0, \sigma^{-2}I)$, $b \in \mathbb{R}^{D'}$ is drawn uniformly from $[0, 2\pi]$, and D' denotes the projected feature dimension. These random projections implicitly approximate an RBF kernel, thereby mapping the original features into a higher-dimensional feature space where nonlinear correlations between modalities become linearly separable [29]. The accumulated covariance matrices $\Sigma_{x'x'}$, $\Sigma_{y'y'}$, and $\Sigma_{x'y'}$ are then used in the same closed-form CCA solution (Eqs. (10)–(16)) to compute the aligned subspace. RFP enriches the representational capacity of CCA without introducing additional trainable parameters or optimization steps, allowing our framework to capture nonlinear visual-textual dependencies.

4.4. Overall Pipeline

During the CL process, we perform a First-Task Training (FTT) stage, where the CLIP model is fine-tuned only on

Table 1. Comparison of CCA-CL with recent CLIP-based CL methods on four benchmarks. PROOF and CLAP adopt real-data replay, whereas SLCA, RAPF, GCL-B and ENGINE perform replay by generating pseudo features. ENGINE and GCL-B are not reproduced because their released code does not provide the required GPT-generated prompts. All datasets are split into 10 tasks.

Method	Exemplar	CIFAR-100		ImageNet-R		ImageNet-100		CUB-200	
		Avg	Last	Avg	Last	Avg	Last	Avg	Last
L2P (2022 CVPR) [38]	✗	81.9	73.0	81.6	75.9	80.5	67.2	70.8	57.9
DualPrompt (2022 ECCV) [37]	✗	81.4	72.5	82.0	75.7	80.6	67.3	69.8	57.4
Continual-CLIP (2022 arXiv) [35]	✗	75.2	66.7	79.1	72.0	85.0	75.4	76.3	71.3
CODAPrompt (2023 CVPR) [33]	✗	76.9	62.2	78.0	67.5	64.1	34.7	73.1	62.9
SLCA (2023 ICCV) [43]	✓	80.5	67.5	75.9	70.3	78.6	59.9	79.1	68.2
MagMax (2024 ECCV) [25]	✗	85.6	79.0	87.1	80.8	86.3	75.9	70.8	62.1
CLAP (2024 NIPS) [18]	✓	78.0	67.5	86.3	80.6	87.3	79.3	83.7	79.0
RAPF (2024 ECCV) [16]	✓	86.1	79.0	85.5	80.2	87.5	80.2	83.0	76.3
ENGINE (2025 ICCV) [48]	✓	86.9	79.2	86.2	80.3	-	-	85.3	79.2
GCL-B (2025 CVPR) [42]	✓	84.4	76.8	-	-	81.8	78.2	84.8	78.5
PROOF (2025 TPAMI) [49]	✓	84.8	76.2	82.8	77.0	84.7	72.4	83.9	79.3
CCA-CL (no FTT)	✗	85.4	81.0	84.7	79.4	86.0	78.0	85.9	79.2
CCA-CL	✗	87.2	83.0	86.8	81.3	86.9	80.2	86.3	79.7

the first task. After this stage, all parameters are frozen for the remaining tasks. This design is motivated not only by the need to balance downstream adaptability and preserve generalization [26, 47], but also by the observation that reducing overall training can further mitigate the asymmetric updates introduced by the AD.

To this end, the overall pipeline is illustrated in Figure 2, CCA-CL first unfreezes all CLIP parameters and performs fine-tuning on the first task to adapt the model to downstream data. After this stage, all parameters of CLIP are frozen, and the model begins to accumulate covariance statistics between visual and textual features. For each subsequent task, CLIP remains frozen and continuously updates the covariance statistics of visual-textual features, followed by computing closed-form CCA projections that maximize the visual-textual correlation. During inference, the frozen CLIP encoders extract image and text features, which are optionally mapped through $\phi_{\text{rfp}}(\cdot)$ and projected into the CCA subspace using the precomputed matrices (A, B) . Classification is performed by computing the similarity between the aligned visual and textual features:

$$\hat{y} = \arg \max_c \frac{(A^\top \phi_{\text{rfp}}(F_v(x)))^\top (B^\top \phi_{\text{rfp}}(z_y^{(c)}))}{\|A^\top \phi_{\text{rfp}}(F_v(x))\|_2 \|B^\top \phi_{\text{rfp}}(z_y^{(c)})\|_2}. \quad (19)$$

The pseudocode of CCA-CL is summarized in Algorithm 1.

5. Experiments

5.1. Experiments Setup

Evaluation Benchmarks. We evaluate our method on four datasets. CIFAR-100 [21] has 100 natural-image classes with 500 training samples per class. ImageNet-R [13] has

200 classes and provides 24,000 images for training and 6,000 images for testing. ImageNet-100 [8] is a 100-class subset of ImageNet-1K; each class comes from the original ImageNet and has about 1,300 training images and 50 validation/test images, which makes it a medium-scale benchmark between CIFAR-100 and full ImageNet. CUB-200 [36] contains 200 bird classes with about 60 images per class, 30 for training and the rest for testing. In this paper, we use a class-incremental setting and randomly split each dataset into 10 disjoint tasks.

Comparison Methods. Our experiments compare two categories of methods: (1) vision-only approaches, including L2P [38], DualPrompt [37], CODA [33], and SLCA [43]; and (2) CLIP-based approaches, including PROOF [49], CLAP [18], RAPF [16], MagMax [25], ENGINE [48], and GCL-B [42]. All methods are deployed with the same CLIP as initialization. Note that ENGINE and GCL-B rely on GPT-generated text prompts to assist training; since the original works did not release the corresponding GPT prompts for some datasets, we do not report reproduced results for those cases.

Implementation details. We implement all methods in PyTorch and conduct experiments on a single NVIDIA RTX 4090 GPU. All methods adopt the same pre-trained CLIP ViT-B/16 as the backbone for fair comparison. In our proposed CCA-CL framework, we perform a light fine-tuning of the CLIP model only on the first task to better adapt it to downstream data [26, 47]. Specifically, we use SGD with a very small learning rate of 1×10^{-6} [43], a batch size of 64, and train for 5 epochs. After this stage, the CLIP network is completely frozen, and all subsequent tasks use the frozen encoder to extract visual and textual features. For the RFP module, we map the features into a

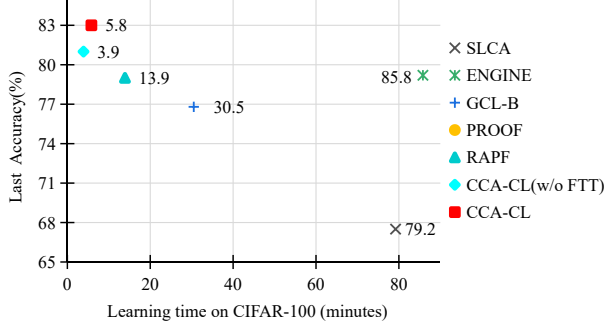


Figure 4. Comparison of learning time and last accuracy on CIFAR-100. CCA-CL achieves the best trade-off between accuracy and training efficiency, requiring only 5.8 minutes while reaching the highest Last Accuracy (83.0%). Its training-free variant, CCA-CL (w/o FTT), further reduces time to 3.9 minutes with only minor performance degradation.

1024 × 6-dimensional space, which provides the best trade-off between performance and memory consumption.

Evaluation metrics. Following the common evaluation protocol [17, 49], we report the average incremental accuracy, which is defined as $\text{Avg.} = \frac{1}{n} \sum_{t=1}^n A_t$, where A_t denotes the accuracy over all seen classes after learning task t . It measures the overall performance throughout the CL process. Additionally, we report the last accuracy corresponds to the accuracy across all categories after the last task, which is denoted as $\text{Last.} = A_n$. It reflects the model’s retained capability at the end of learning.

5.2. Comparison Results

Table 1 presents a comparison of our approach with recent CLIP-based CL methods. Across most datasets, our method consistently outperforms all others, including those relying on replay or GPT-generated prompts. Specifically, in terms of Last Accuracy, CCA-CL achieves the best performance on all datasets, especially on CIFAR-100, outperforming the second-best method by up to 3.8%. Regarding Avg Accuracy, CCA-CL achieves the highest results on all benchmarks except ImageNet-100, where CLAP and RAPF perform slightly better, both relying on replay mechanisms. We do not rely on replay or GPT-generated text, yet CCA-CL still outperforms all non-replay methods in Avg Accuracy.

In addition, we also report the performance of CCA-CL without first-task training in the second-to-last row. In this variant, the model performs no parameter updates and thus serves as a training-free method, offering fast execution. Under this constraint, CCA-CL (no first-task training) still achieves competitive performance—for example, ranking second on CIFAR-100 and outperforming most competing methods on other datasets. This confirms that explicitly realigning the visual-textual subspace through CCA, further enhanced by Random Fourier Projection (RFP), effectively strengthens the visual-textual correlation, thus en-

abling better performance with minimal training effort.

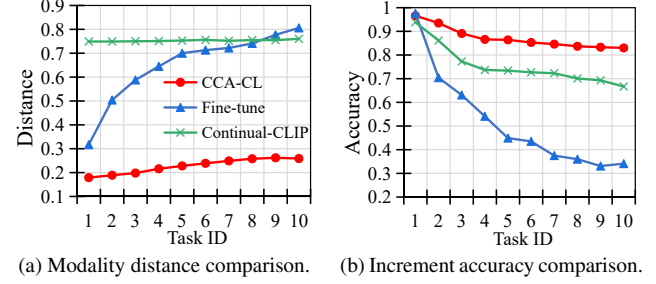


Figure 5. Comparison of modality distance and increment accuracy. (a) CCA-CL maintains a small and stable distance by projecting features into a correlated subspace. (b) CCA-CL consistently achieves the best accuracy by reducing and stabilizing the modality distance.

5.3. Motivation Analysis

We further investigate the effect of modality distance on CL performance by comparing CCA-CL, Fine-tune, and Continual-CLIP baselines. As shown in Figure 5, directly fine-tuning CLIP leads to progressively increasing modality distance due to asymmetric drift, which in turn causes a continuous drop in accuracy. On the other hand, Continual-CLIP avoids such drift by freezing all CLIP parameters, keeping the distance stable. However, since no adaptation is performed, the modality distance remains large throughout, limiting the overall performance. In contrast, our CCA-CL approach identifies a shared subspace using CCA that minimizes the distance between visual and textual features. This not only reduces the initial modality gap but also suppresses further drift across tasks. As a result, CCA-CL consistently achieves lower distance and higher accuracy, demonstrating its effectiveness in mitigating asymmetric drift and preserving robust vision-language alignment.

5.4. Efficiency Analysis

Figure 4 illustrates the trade-off between training efficiency and accuracy on CIFAR-100. We represent the total time required to complete all incremental tasks. Our proposed CCA-CL completes all tasks in only 5.8 minutes, which is substantially faster than other state-of-the-art approaches. For example, RAPF requires 13.9 minutes, which is more than twice as long, while ENGINE and SLCA take over 80 minutes, making them more than ten times slower. Despite this significant reduction in training time, CCA-CL still achieves the highest accuracy (83.0%), demonstrating an excellent balance between efficiency and performance. Furthermore, when the first-task training stage is removed (CCA-CL w/o FTT), the total runtime further decreases to just 3.9 minutes. Although the accuracy slightly drops to 81.0%, this variant remains highly competitive, outperforming other methods while requiring minimal learning time.

Table 2. Ablation results on CIFAR-100. Each component—CCA, RFP, and FTT—contributes to improved performance, with the full framework achieving 83.0% accuracy.

CCA	RFP	FTT	Accuracy
			66.7
✓			76.5
✓	✓		81.0
✓	✓	✓	83.0

5.5. Ablation Study

We conduct ablation experiments on CIFAR-100 to evaluate the contributions of each component in our proposed framework. As shown in Table 2, our baseline is Continual-CLIP [35], which freezes the CLIP model entirely and does not include any additional trainable parameters. This setup achieves a base accuracy of 66.7%. Introducing CCA significantly improves the performance to 76.5% (+9.8), as it finds a correlated subspace for visual and textual features and minimizes the modality distance. When combined with RFP, which captures nonlinear dependencies that linear CCA cannot model, the performance further increases to 81.0%, yielding an additional gain of 4.5%. Finally, enabling First-Task Tuning (FTT) allows CLIP to better adapt to the downstream domain while preserving its generalization and zero-shot capabilities. This provides a performance boost, achieving the best accuracy of 83.0%.

5.6. Parameter Efficiency

Effect of Width W . To investigate the influence of the RFP dimensionality, we vary the output width W from 2048 to 10240 and report the corresponding last accuracy (%) and memory consumption (G) in Table 3. As the projection width increases, both accuracy and memory usage rise, revealing a trade-off between representational capacity and computational efficiency. A larger W enables richer nonlinear feature mapping and improves the expressiveness of the projected subspace, benefiting the CCA alignment by capturing more fine-grained correlations between visual and textual features. When $W=6144$, CCA-CL achieves the optimal balance, reaching the highest accuracy of 81.0% with a moderate memory footprint of only 6.3 GB. Beyond this point, increasing W provides negligible accuracy gains while significantly raising memory consumption.

Effect of Energy Threshold η . We evaluate the sensitivity of CCA to the energy threshold η under the setting without First-Task Tuning (FTT) and with RFP ($W = 6144$). The parameter η determines the proportion of cumulative canonical correlation energy to retain, which in turn controls the number of canonical directions selected. The results are shown in Table 4. In our setup, when $\eta = 1.0$, all channels are retained for each task, resulting in $\bar{r} = 6144$. In contrast, for $\eta < 1.0$, the number of retained directions is

Table 3. Parameter sensitivity analysis of the RFP module on CIFAR-100. The setting of 6144 achieves the best trade-off between accuracy and memory usage.

W	2048	4096	6144	8192	10240
Memory(G)	3.1	4.3	6.3	8.8	11.9
Last Acc(%)	79.7	80.3	81.0	80.6	80.2

Table 4. Sensitivity of the CCA energy threshold η on accuracy and subspace size without FTT. $\bar{r}$ denotes the average number of retained canonical directions across tasks. W in RFP set to 6144.

η	0.9	0.95	0.99	0.999	1
Last Acc(%)	76.5	80.3	81.0	80.9	80.8
$\bar{r}$	34.0	42.5	47.2	48.5	6144

significantly smaller. For example, at $\eta = 0.99$, the average retained dimension $\bar{r}$ is only 47.2. As η increases, accuracy gradually improves but saturates around $\eta = 0.99$. Beyond this point, further increases lead to marginal or even negative performance gains. This is because retaining too many components may introduce noise into the projection space. Overall, $\eta = 0.99$ provides the best trade-off between compact subspace representation and model performance.

6. Conclusion

In this paper, we identify an overlooked issue in CLIP-based continual learning: Asymmetric Drift, which arises from unequal shifts in the visual and textual feature distributions across tasks. This asymmetry leads to imbalanced changes between the visual and text encoders, causing their features to diverge over time. As a result, the modality distance grows throughout the continual learning process, ultimately degrading cross-modal alignment and continual learning performance. To address this issue, we propose a statistically grounded solution based on Canonical Correlation Analysis. By solving the canonical correlation equation, our method identifies a shared subspace where the distance between visual and textual features is minimized, effectively mitigating the Asymmetric Drift problem. What’s more, to overcome the limitation of linear modeling in CCA, we further incorporate Random Fourier Projection to capture nonlinear dependencies and enhance representational capacity. While CCA-CL achieves superior accuracy and efficiency, it still relies on first-task fine-tuning of CLIP. In the future, we plan to explore integrating parameter-efficient tuning strategies to further improve scalability and stability.

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